Day 3.2 Progress Report

WorldSkills Module A - Internal DNS, DHCP, DDNS, and Troubleshooting Log

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Lab platform: Proxmox VE, OPNsense, Debian int-srv01, Ubuntu client
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Purpose

This document summarizes the work completed from the beginning of the Day 3.2 session. It captures what was configured, what was verified, what failed, what was learned, and what should be continued next. It is written as a short engineering record for GitHub or portfolio documentation.

> Key conclusion: the module expects DHCP and DDNS to be implemented on **int-srv01**, while the lab initially used OPNsense Kea DHCP. OPNsense was useful for learning and troubleshooting, but the scoring architecture should move DHCP/DDNS to the Debian internal server.

1. Official Requirement Interpreted

The Module A requirement places Internal DNS and DHCP with DDNS under **int-srv01.int.worldskills.ph**. The internal server is expected to provide authentication, CA, file, DNS, DHCP/DDNS, and syslog services. The firewall is a separate role and later acts as router, firewall, proxy, VPN endpoint, and DHCP relay.

Area	Required by module	Implication for lab
Internal DNS	BIND9 on int-srv01 with int.worldskills.ph and reverse zones	Keep BIND9 on Debian int-srv01
DHCP with DDNS	DHCP server assigns IPv4/IPv6 and registers int-client via DDNS	Move DHCP/DDNS from OPNsense to int-srv01
Firewall	Packet forwarding, NAT, WireGuard, proxy, DHCP relay to int-srv01	OPNsense/Firewall should not be the scoring DHCP server

2. Network and VM Context

The lab used Proxmox VE as the virtualization platform. OPNsense was installed as the firewall/router, Debian acted as int-srv01, and Ubuntu acted as the client used for DHCP and DNS testing.

```
Current main lab roles:
- OPNsense firewall: 10.1.10.1/24 on INT/LAN
- Debian int-srv01: 10.1.10.10/24
- Ubuntu client: DHCP-assigned address, observed as 10.1.10.100/24 later
- Internal DNS zone: int.worldskills.ph
- Internal IPv4 reverse zone: 10.1.10.in-addr.arpa
```

3. OPNsense Rebuild and Interface Work

OPNsense was reset/reinstalled to review the firewall setup from the beginning. Multiple VirtIO interfaces were detected as vtnet0, vtnet1, vtnet2, and vtnet3. Interface assignment was reviewed and the firewall was brought back to a stable state with WAN and LAN/INT connectivity.

```
Important OPNsense lessons:
- vtnet interfaces are VM NICs created by Proxmox.
- vtnet0 was mapped to WAN based on the Proxmox NIC order.
- LAN/INT was configured as 10.1.10.1/24.
- OPNsense could ping 8.8.8.8 and resolve google.com.
- DNSmasq DNS & DHCP caused conflicts and was disabled.
```

4. BIND9 DNS Foundation on Debian

BIND9 was installed on Debian and verified as running. The server listened on port 53 and answered authoritative queries for the internal zone.

```
apt install bind9 bind9-utils dnsutils -y
systemctl status named
ss -tulnp | grep named
```

4.1 Forward Zone

A forward zone was created for the official internal namespace **int.worldskills.ph**.

```
// /etc/bind/named.conf.local
include "/etc/bind/ddns.key";

zone "int.worldskills.ph" {
    type master;
    file "/etc/bind/db.int.worldskills.ph";
    allow-update { key "ddns-key"; };
};

zone "10.1.10.in-addr.arpa" {
    type master;
    file "/etc/bind/db.10.1.10";
};
```

```
// /etc/bind/db.int.worldskills.ph
$TTL 604800
@ IN SOA ns1.int.worldskills.ph. admin.int.worldskills.ph. (
    2026052901
    604800
    86400
    2419200
    604800 )

@           IN NS    ns1.int.worldskills.ph.
ns1         IN A     10.1.10.10
int-srv01   IN A     10.1.10.10
@           IN A     10.1.10.10
_auth._tcp  IN SRV   10 50 389 int-srv01.int.worldskills.ph.
```

4.2 Reverse Zone

```
// /etc/bind/db.10.1.10
$TTL 604800
@ IN SOA ns1.int.worldskills.ph. admin.int.worldskills.ph. (
    2026052901
    604800
    86400
    2419200
    604800 )

@  IN NS  ns1.int.worldskills.ph.
10 IN PTR int-srv01.int.worldskills.ph.
```

4.3 Validation Commands

```
named-checkconf
named-checkzone int.worldskills.ph /etc/bind/db.int.worldskills.ph
named-checkzone 10.1.10.in-addr.arpa /etc/bind/db.10.1.10
systemctl reload named

dig @10.1.10.10 ns1.int.worldskills.ph
dig @10.1.10.10 int-srv01.int.worldskills.ph
dig @10.1.10.10 -x 10.1.10.10
```

Forward and reverse lookups for int-srv01 were confirmed working.

5. LDAP SRV Record Review

A service discovery record was created and tested. The first attempt accidentally created a duplicated suffix because the target hostname did not have a trailing dot. This was corrected.

```
Wrong result observed:
int-srv01.int.worldskills.ph.int.worldskills.ph.

Correct SRV target:
int-srv01.int.worldskills.ph.

dig @10.1.10.10 SRV _auth._tcp.int.worldskills.ph

Correct answer observed:
_auth._tcp.int.worldskills.ph. 604800 IN SRV 10 50 389 int-srv01.int.worldskills.ph.
Additional section returned:
```

```
int-srv01.int.worldskills.ph. 604800 IN A 10.1.10.10
```

Note: the marking sheet later showed the expected record as **_ldap._tcp.auth.int.worldskills.ph**. This should be added next to avoid losing marks due to exact label mismatch.

6. DHCP on OPNsense - What Worked and What Was Learned

OPNsense Kea DHCPv4 was configured first. It successfully assigned IPv4 leases and distributed DNS options to the Ubuntu client. This was technically useful, but later we discovered that the module expects DHCP/DDNS under int-srv01.

```
OPNsense Kea DHCPv4 subnet values used:
Subnet:      10.1.10.0/24
Pool:        10.1.10.100 - 10.1.10.200
Gateway:     10.1.10.1
DNS server:  10.1.10.10
Domain:      int.worldskills.ph

Client validation commands:
resolvectl dns
resolvectl domain
sudo dhclient -r ens18
sudo dhclient -v ens18
host int-srv01.int.worldskills.ph 10.1.10.10
```

Dnsmasq DNS & DHCP was discovered to be enabled and interfered with Kea DHCP, causing clients to receive DNS = 10.1.10.1 instead of DNS = 10.1.10.10. Disabling Dnsmasq fixed DHCP DNS option delivery.

7. DDNS Work and Troubleshooting

DDNS was started using OPNsense Kea DHCP and Debian BIND. A TSIG key was generated manually using OpenSSL because dnssec-keygen did not support HMAC algorithms in this installation and tsig-keygen was unavailable.

```
openssl rand -base64 32

// /etc/bind/ddns.key
key "ddns-key" {
    algorithm hmac-sha256;
    secret "***REDACTED***";
};
```

The key was included in BIND and allow-update was added to the int.worldskills.ph zone.

```
include "/etc/bind/ddns.key";

zone "int.worldskills.ph" {
    type master;
    file "/etc/bind/db.int.worldskills.ph";
    allow-update { key "ddns-key"; };
};
```

7.1 OPNsense DDNS Agent Issue Fixed

The OPNsense DDNS Agent initially failed because it was configured to bind to 10.1.10.10, which belongs to Debian, not OPNsense. It was corrected to listen locally.

```
Original issue:
DHCP_DDNS_QUEUE_MGR_START_ERROR
Can't assign requested address
DHCP_DDNS_NOT_ON_LOOPBACK

Corrected DDNS Agent setting:
Bind address: 127.0.0.1
Port: 53001
```

7.2 Kea DDNS Config Confirmed

The generated OPNsense Kea configuration confirmed that DHCP was set to send DDNS updates to the local DDNS Agent.

```
"ddns-qualifying-suffix": "int.worldskills.ph",
"ddns-send-updates": true,

"dhcp-ddns": {
    "enable-updates": true,
    "server-ip": "127.0.0.1",
    "server-port": 53001
}
```

The DDNS Agent config also showed a forward-ddns domain pointing to 10.1.10.10 with the TSIG key.

7.3 Manual nsupdate Test

A manual nsupdate test was performed to isolate whether the failure was on OPNsense or BIND. The request reached BIND and TSIG was accepted, but BIND returned SERVFAIL.

```
nsupdate -d -k /etc/bind/ddns.key
server 10.1.10.10
update add test.int.worldskills.ph. 300 A 10.1.10.123
send

Observed result:
Reply from update query: status: SERVFAIL
TSIG: NOERROR
```

This indicates that the key and network path are valid. The remaining failure is likely inside BIND update processing, such as journal file creation, directory permissions, AppArmor policy, or zone update handling.

7.4 Permission Check

BIND runs as user bind. Testing showed that user bind cannot create files in /etc/bind. No journal file was created.

```
sudo -u bind touch /etc/bind/test-bind-file
Result: Permission denied

ls -l /etc/bind/*.jnl
Result: No such file or directory
```

> Potential fix to test later: allow BIND to create the .jnl journal file, or move dynamic zones to a writable directory such as /var/lib/bind and update named.conf.local accordingly. Moving dynamic zones to /var/lib/bind is usually cleaner than making all of /etc/bind writable.

8. Marking Sheet Mapping

A2 Marking Item	Current Status	Notes
DNS: A, AAAA and PTR for int-srv01.int.worldskills.ph exists	Partial	A and PTR exist. AAAA still needs verification/addition.
DNS: SRV Record _ldap._tcp.auth.int.worldskills.ph exists	Partial	A SRV record exists, but exact marking name may differ. Add _ldap._tcp.auth.
DNS: Server accepts recursive queries	Done	google.com resolution succeeded.
DNS: int-srv01 is secondary NS for dmz.worldskills.ph	Not started	Requires DMZ DNS servers first or simulated zone transfer.
DNS: int-srv01 secondary NS for DMZ reverse zones	Not started	Requires 10.1.20.0/24 and IPv6 reverse slave zones.
DHCP leases assign IPv4 and IPv6 to int-client	Not compliant yet	OPNsense provided IPv4, but module expects int-srv01 DHCP and IPv6.
Reverse DNS created for DHCP-assigned IP	Not finished	Depends on DDNS working.
DDNS dynamic updates for IPv4 configured	In progress	TSIG and configs exist; BIND returns SERVFAIL on nsupdate.
int-client receives IPv4/IPv6 and can ping INT/DMZ servers	Not started	Need int-client network and IPv6 routing/DHCPv6.

9. Major Lessons Learned

• DNS record targets need trailing dots when using FQDNs inside zone files.

• Dnsmasq and Kea DHCP can conflict if both are enabled on the same interface.

• A working DNS query path does not mean dynamic updates are working.

• TSIG NOERROR means the key was accepted; SERVFAIL means the server had an internal update-processing issue.

• For WorldSkills, correct service placement matters. Functional services in the wrong host may not score.

• The firewall should likely relay DHCP to int-srv01 rather than act as the DHCP server for this module.

10. Recommended Next Steps

Priority	Task	Reason
1	Stop using OPNsense as the scoring DHCP server	Module places DHCP/DDNS under int-srv01.
2	Install and configure DHCP on Debian int-srv01	Required for A2 scoring.
3	Move dynamic BIND zone file to /var/lib/bind or fix journal permissions safely	Manual nsupdate currently returns SERVFAIL.
4	Add exact SRV record _ldap._tcp.auth.int.worldskills.ph	Matches marking sheet.
5	Add AAAA record for int-srv01	Required in marking sheet.
6	Configure DHCPv6 scope 2001:db8:1001:40::100-199/64	Required by DHCP section.
7	Configure firewall DHCP relay to int-srv01	Matches firewall marking item.

11. Suggested Starting Point for Next Session

Begin by making BIND dynamic updates work locally on Debian before configuring Kea DHCP on int-srv01. A safe approach is to place dynamic zone files in /var/lib/bind.

```
Proposed next test path:
1. Copy db.int.worldskills.ph to /var/lib/bind/db.int.worldskills.ph
2. chown bind:bind /var/lib/bind/db.int.worldskills.ph
3. Update named.conf.local file path to /var/lib/bind/db.int.worldskills.ph
4. reload named
5. Run nsupdate again
6. Verify test.int.worldskills.ph resolves
7. Confirm .jnl file is created
8. Then install/configure DHCP/DDNS on int-srv01
```

End of report